

# Emergence of seasonal influenza A(H3N2) variants with immune escape potential warrants enhanced molecular and epidemiological surveillance for the 2025–2026 season

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**BACKGROUND:** All of the major antigenic changes in influenza A(H3N2) viruses since 1968 have involved mutations at just nine amino acid positions, called cluster transition sites, surrounding the receptor binding region of the hemagglutinin surface protein. During the Northern Hemisphere (NH) 2024–2025 influenza season, A(H3N2) variants emerged with multiple parallel substitutions affecting cluster transition sites 135, 145, 158, and/or 189, with implications for the 2025–2026 season.

**METHODS:** Using >24,000 global A(H3) sequences between September 2024 and August 2025, we assessed the nature and frequency of amino acid mutations among emerging NH and Southern Hemisphere (SH) A(H3N2) variants relative to the 2024–2025 subclade J and updated 2025–2026 subclade J.2 vaccine reference strains. We contextualized based on historic amino acid variation among >210,000 global A(H3) sequences since 1968 and publicly available antigenic characterization data relative to 2024–2025 and 2025–2026 vaccine reference strains.

**RESULTS:** Virtually all circulating A(H3N2) viruses in 2024–2025 were subclade J.2. In Europe, about one-third had cluster transition site mutations, mostly T135A with S145N. In North America, more than two-thirds had cluster transition site mutations, including T135K or S145N, with a late-season increase in doubly mutated J.2.3 (N158K + K189R) and J.2.5 (S145N + N158K) subclades. The SH 2025 season showed an increase in J.2.3 and emergence of J.2.4 (T135K + K189R), including a further drifted J.2.4.1 variant with additional N158D and other mutations, recently renamed subclade K. A substantial proportion of J.2.3 and J.2.4 viruses are antigenically distinct from the 2025–2026 influenza vaccine.

**CONCLUSION:** Influenza A(H3N2) variants with a combination of cluster transition site mutations emerged during the NH 2024–2025 season. A further drifted and vaccine-mismatched variant, called subclade K, arose during the SH 2025 season and is projected to predominate among A(H3N2) viruses for the NH 2025–2026 season. While mismatched vaccines can still provide protection, enhanced genetic, antigenic, and epidemiological (eg, vaccine effectiveness, disease burden) monitoring are warranted to inform risk assessment and response.

**KEYWORDS:** A(H3N2), amino acids, antigenic drift, escape variants, genetic drift, influenza, risk assessment

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**HISTORIQUE :** Toutes les modifications antigéniques majeures aux virus de l'influenza A(H3N2) depuis 1968 découlent des mutations de neuf positions d'acides aminés seulement qui entourent la région des récepteurs de liaison de l'hémagglutinine, une protéine de surface. Pendant la saison 2024–2025 de l'influenza dans l'hémisphère Nord (HN), des variants A(H3N2) ont émergé, dotés de multiples substitutions parallèles qui touchent les positions 135, 145, 158 ou 189, ce qui a des conséquences pour la saison 2025–2026 de l'influenza.

**MÉTHODOLOGIE :** Entre septembre 2024 et août 2025, les auteurs ont utilisé plus de 24 000 séquences A(H3) globales pour évaluer la nature et la fréquence des mutations d'acides aminés dans les variants A(H3N2) émergents de l'HN et de l'hémisphère Sud (HS) par rapport à celles des souches référence vaccinale du sous-clade J de 2024–2025 et du sous-clade J.2 de 2025–2026. Ils les ont contextualisées d'après la variation historique d'acides aminés de plus de 210 000 séquences d'A(H3) globales depuis 1968 et les données de caractérisation antigénique accessibles au public relatives aux souches de référence vaccinales de 2024–2025 et de 2025–2026.

**RÉSULTATS :** Pratiquement tous les virus A(H3N2) en circulation en 2024–2025 faisaient partie du sous-clade J.2. En Europe, environ le tiers comportait des mutations à diverses positions, notamment T135A et S145N. En Amérique du Nord, plus des deux tiers étaient dotés de mutations à diverses positions, y compris T135K ou S145N, et présentaient une augmentation des sous-clades J.2.3 (N158K+K189R) et J.2.5 (S145N+N158K) qui ont subi une double mutation en fin de saison. La saison 2025 en HS a révélé une augmentation du sous-clade J.2.3 et l'émergence du sous-clade J.2.4 (T135K+K189R), y compris un variant du sous-clade J.2.4.1 qui a subi une dérive plus marquée incluant une mutation supplémentaire à la position N158D et d'autres mutations. Ce variant a récemment reçu le nom de sous-clade K. Une proportion substantielle des virus aux sous-clades J.2.3 et J.2.4 sont distincts sur le plan antigénique de ceux contenus dans le vaccin contre l'influenza pour la saison 2025–2026.

**CONCLUSIONS :** Des variants de l'influenza A(H3N2) dotés d'une combinaison de mutations aux diverses positions ont émergé pendant la saison 2024–2025 de l'HN. Un variant ayant subi une dérive génétique plus importante et qui ne concorde pas au vaccin, désormais nommé sous-clade K, a émergé pendant la saison 2025 en HS et devrait prédominer parmi les virus A(H3N2) pour la saison 2025–2026 en HN. Bien que des vaccins non concordants puissent offrir une certaine protection, une surveillance génétique, antigénique et épidémiologique des vaccins (p. ex., leur efficacité) s'impose pour mieux évaluer le risque et la réponse s'y rapportant.

**MOTS-CLÉS :** influenza, A(H3N2), dérive génétique, dérive antigénique, échappement immunitaire des variants, évaluation du risque, acides aminés

## LAY SUMMARY

After a period of minimal change following the COVID-19 pandemic, influenza A(H3N2) viruses started to evolve in important ways during the Northern Hemisphere's 2024–2025 influenza season. During the Southern Hemisphere's 2025 influenza season, a further evolved and vaccine-mismatched A(H3N2) variant arose and is projected to predominate during the Northern Hemisphere's 2025–2026 influenza season. While mismatched vaccines can still provide protection against circulating variants, enhanced surveillance is warranted, including timely influenza virus sequencing, antigenic characterization, and vaccine effectiveness monitoring to inform risk assessment and response.

## INTRODUCTION

During the 2024–2025 Northern Hemisphere (NH) influenza season, both influenza A (H1N1)pmd09 and H3N2 subtypes co-circulated, with regional differences in their relative contribution (1). The A(H1N1)pdm09 subtype

dominated overall, but regions like North America, Europe, and the Eastern Mediterranean also experienced substantial A(H3N2) co-circulation (1–3). In North America, the 2024–2025 dual influenza A epidemics were associated with increased severity affecting all age groups, ultimately rivaling the severity of any other influenza season of the past decade, as shown by surveillance indicators such as cumulative and peak week influenza hospitalization rates and paediatric deaths (4–6).

In early work to antigenically characterize A(H3N2) viruses between 1968 and 2003, Smith et al identified 11 antigenically distinct variant clusters, with transition between clusters associated with major immunological escape requiring vaccine update (7). Koel et al subsequently attributed all such antigenic cluster transitions to mutations at just nine mapped amino acid positions (or residues) on the globular head of the hemagglutinin (HA) surface protein (8). These nine pivotal amino acid positions are located around the receptor binding site (RBS) within exposed regions of antigenic sites A and B (9). Historic mutations at seven of the amino acid positions were directly involved in cluster transitions (ie, cluster transition sites), including

one position in antigenic site A (145) and six positions in antigenic site B (155, 156, 158, 159, 189, and 193) (8). Two accessory positions in antigenic site A (133 and 135) also played a facilitating role in immune escape and cluster transition (8). All other major antigenic cluster transitions between 2003 and 2018 were similarly driven by mutations in these same nine residues (10,11). Because they may herald a potentially important impact on antigenicity, immune escape, and/or vaccine effectiveness (VE), these nine amino acid positions require close genetic and antigenic monitoring among circulating viruses. Mutations independently affecting the same amino acid positions but within different phylogenetic branches (ie, parallel substitutions) suggest some sort of fitness advantage may be conferred to viruses accruing them (12).

After several years of only limited genetic or antigenic change following the COVID-19 pandemic, A(H3N2) viruses have shown recent signs of emerging from their genetic bottleneck, with multiple parallel substitutions affecting several cluster transition and/or accessory residues, namely positions 135, 145, 158, and 189, alone or in combination (13). To address antigenically significant evolution, the World Health Organization (WHO) routinely convenes in February each year to select updated influenza vaccine strains for the NH, but circulating viruses and their relative contribution can continue to evolve thereafter (2). Such evolution continued among circulating A(H3N2) viruses subsequent to the WHO's February 2025 updated vaccine recommendation for the NH 2025–2026 season, leading the WHO in September 2025 to recommend a further update to the A(H3N2) vaccine component for the Southern Hemisphere (SH) 2026 influenza season (3). However, the latter recommendation comes too late for incorporation in the NH 2025–2026 influenza vaccination campaign. In this analysis, we use publicly available sequencing and antigenicity data to summarize the evolving A(H3N2) landscape with the aim of enhancing awareness, stimulating strategic molecular and epidemiological surveillance, and informing public health risk assessment and response entering the NH 2025–2026 influenza season.

## METHODS

### Molecular surveillance data

We downloaded all worldwide influenza A(H3N2) HA sequences available within the Global Initiative on Sharing All Influenza Data (GISAID), specifically those with collection dates spanning January 1, 1968 to August 30, 2025 and uploaded by September 15, 2025 (14), aligning these

using MAFFT v7.245 (15). Sequence data were deduplicated based on sequence name with original specimen sequences retained over cell-passaged and the latter over egg-passaged isolates.

### Global mutational frequencies

We analyzed mutational frequencies during the NH 2024–2025 and SH 2025 seasons using A(H3N2) sequences with specimen collection dates between September 1, 2024 (week 36), and August 30, 2025 (week 35). We used Nextclade (August 20, 2025 version) for mutation identification and clade/subclade assignment (16). Because they comprised virtually all downloaded A(H3N2) sequences, we summarized only for A(H3N2) clade 2a.3a.1 viruses (13), also known as subclade J or its descendants based upon Nextstrain nomenclature (17). A key to Nextstrain nomenclature and defining substitutions for subclade J descendant strains is provided in Table 1.

We identified variant-specific mutations in relation to the SH 2024 and NH 2024–2025 cell-passaged A(H3N2) clade 2a.3a.1 (subclade J) vaccine strain, A/Massachusetts/18/2022 (2). For simplicity, we refer to circulating viruses as “wildtype” if they have no mutations at any of the nine antigenic cluster transition or accessory positions relative to A/Massachusetts/18/2022, regardless of other subclade designations. At first mention, we annotate noteworthy characteristics of mutations in parentheses, such as when involving an antigenic site (A–E) (9), the RBS, and gain or loss of glycosylation (+CHO/–CHO).

### Antigenic characterization

We summarize publicly available antigenic characterization data based upon hemagglutination inhibition (HI) assay findings extracted from the most recent February and September 2025 interim reports of the Francis Crick Institute Worldwide Influenza Centre (13) and the September 2025 report of the WHO SH 2026 vaccine strain selection meeting (3). Per convention, we define antigenic distinction by  $\geq 8$ -fold reduction in HI titre against the circulating test variant when compared to the homologous vaccine strain against which ferret antisera were raised (18). Fold reductions in HI titre can also be represented by antigenic distance (AD) units of change, with the following relationship:  $\times$  AD units =  $2^x$  fold reduction (eg, 1, 2, 3, 4, or 5 AD units of change correspond with 2-fold, 4-fold, 8-fold, 16-fold, or 32-fold reductions, respectively).

We summarize fold-reductions relative to representative NH 2024–2025 (subclade J) and upcoming NH 2025–2026 (subclade J.2) vaccine strains (the latter also shared with

**Table 1:** Influenza A(H3N2) clade 2a.3a.1 (J subclades), defining substitutions and tallies (N = 24,261)

| J Subclade (n; %)*       | Defining HA1 substitutions <sup>†</sup>                                                         | Additional nondefining HA1 substitutions in cluster transition positions <sup>†,‡</sup>                                                                                                                                                                                               | Wildtype (W) or non-wildtype (NW) <sup>§</sup> | Variant tallies (n; %)*                                        |
|--------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------------------|
| J (30; <1)               | Reference                                                                                       | - <sup>¶</sup><br>+ <b>S145N (A)</b><br>+ <b>T135A (A) (RBS) (-CHO)</b>                                                                                                                                                                                                               | W<br>NW<br>NW                                  | 25 (<1)<br>3 (<1)<br>2 (<1)                                    |
| J.1 (20; <1)             | J + I25V**                                                                                      | -<br>+ <b>S145N<sup>††</sup></b><br>+ <b>N159Y (B)</b>                                                                                                                                                                                                                                | W<br>NW<br>NW                                  | 16 (<1)<br>3 (<1)<br>1 (<1)                                    |
| J.1.1 (213; <1)          | J.1 + <b>S145N (A)<sup>‡‡</sup></b>                                                             | -<br>+ <b>N158D (B)</b> or <b>K189R (B)</b> or <b>T135K (A) (RBS) (-CHO)</b> or <b>N133D (A) (-CHO)</b>                                                                                                                                                                               | NW<br>NW                                       | 206 (<1)<br>7 (<1)                                             |
| J.2 (16,960; 70)         | J + N122D (A) (-CHO) + K276E (C)                                                                | -<br>+ <b>T135K</b><br>+ <b>T135A + S145N</b><br>+ <b>S145N<sup>§§</sup></b><br>+ <b>N133D/H/S/T</b> and/or <b>T135A/K</b> and/or <b>S145N/G/R</b> and/or <b>T155A (B)(RBS)</b> and/or <b>S156H (B)</b> and/or <b>N158H/K/S/T/D</b> and/or <b>K189R/N/S</b> and/or <b>S193F/T (B)</b> | W<br>NW<br>NW<br>NW<br>NW                      | 7,848 (32)<br>4,115 (17)<br>1,357 (6)<br>3,045 (13)<br>595 (2) |
| J.2.1 (712; 3)           | J.2 + F79L + P239S                                                                              | -<br>+ <b>T135K</b> only<br>+ <b>N133S</b> or <b>T135A</b> or <b>S145N/G</b> or <b>N158K</b> or <b>N159S</b> or <b>K189R</b>                                                                                                                                                          | W<br>NW<br>NW                                  | 622 (3)<br>46 (<1)<br>44 (<1)                                  |
| J.2.2 (4,258; 18)        | J.2 + S124N (A)                                                                                 | -<br>+ <b>S145N</b><br>+ <b>N133D</b> and/or <b>T135K/A/I</b> and/or <b>S145N/G/K/R</b> and/or <b>T155S</b> and/or <b>S156H</b> and/or <b>N158K</b> and/or <b>N159S</b> and/or <b>K189R</b> and/or <b>S193F</b>                                                                       | W<br>NW<br>NW                                  | 3,276 (14)<br>834 (3)<br>148 (<1)                              |
| J.2.3 (501; 2)           | J.2 + <b>N158K (B) + K189R (B)<sup>***</sup></b>                                                | -<br>+ <b>S145N/G</b>                                                                                                                                                                                                                                                                 | NW<br>NW                                       | 466 (2)<br>35 (<1)                                             |
| J.2.4 (309; 1)           | J.2 + <b>T135K (A) (RBS) (-CHO) + K189R (B)</b>                                                 | - <sup>†††</sup><br>+ <b>K135N (-CHO)</b> or <b>S145N</b> or <b>N158K</b>                                                                                                                                                                                                             | NW<br>NW                                       | 301 (1)<br>8 (<1)                                              |
| J.2.4.1 (now K) (45; <1) | J.2.4 + K2N + S144N (A) (+CHO) + <b>N158D</b> + I160K (B) + Q173R (D) + T328A <sup>***§§§</sup> | -                                                                                                                                                                                                                                                                                     | NW                                             | 45 (<1)                                                        |
| J.2.5 (1,197; 5)         | J.2 + D104N + <b>S145N + N158K</b>                                                              | -<br>+ <b>T135A</b> or <b>N145D</b> or <b>N156H</b> or <b>N159S</b> or <b>K189R</b>                                                                                                                                                                                                   | NW<br>NW                                       | 1,183 (5)<br>14 (<1)                                           |
| J.3 (3; <1)              | J <sup>†††</sup>                                                                                | -                                                                                                                                                                                                                                                                                     | W                                              | 3 (<1)                                                         |
| J.4 (13; <1)             | J + Q173R (D) + K276E (C)                                                                       | + <b>K189R</b>                                                                                                                                                                                                                                                                        | NW                                             | 13 (<1)                                                        |



**Table 1:** Continued (abbreviations, footnotes)

Sequences identified worldwide within the Global Initiative on Sharing All Influenza Data (GISAID) platform between September 1, 2024 (week 36), and August 30, 2025 (week 35) (14). Nomenclature and subclade defining substitutions as per Nextstrain (17). Only HA1 defining substitutions listed in the table; antigenic cluster transition site substitutions **bolded**; hemagglutinin stalk (HA2) substitutions footnoted. All substitutions are in relation to the cell-based A/Massachusetts/18/2012 vaccine reference strain (clade 2a.3a.1, subclade J)

\*Column percent as a fraction of the total number of A(H3N2) clade 2a.3a.1 sequences included (24,261)

†Antigenic sites and other details such as involvement of the RBS and +/− CHO shown in parentheses at first mention by column only

‡Additional cluster transition site substitutions are listed if present in ≥5% of specified subclade, otherwise displayed combined in separate row

§Wildtype defined by no amino acid mutations at any of the nine antigenic cluster transition or accessory sites; non-wildtype defined by one or more such amino acid mutations, itemized in **bold**

¶A(H3N2) vaccine reference strains for Northern Hemisphere 2024–2025 and Southern Hemisphere 2024 influenza season, including A/Massachusetts/18/2012 (cell-based) and A/Thailand/08/2022 (egg-based)

\*\*HA2 defining substitution: V18M

††Lacks HA2 defining substitution: V89I, characteristic of J.1.1

‡‡HA2 defining substitution: V89I

§§Influenza A(H3N2) vaccine reference strains for Northern Hemisphere 2025–2026 and Southern Hemisphere 2025 influenza season, including A/District of Columbia/27/2023 (cell-based) and A/Croatia/10136RV/2023 (egg-based), both with additional S145N (A) substitution

\*\*\*HA2 defining substitution: S49N

†††Influenza A(H3N2) vaccine reference strains for Southern Hemisphere 2026 vaccine, including A/Sydney/1359/2024 (cell-based) and A/Singapore/GP20238/2024 (egg-based)

‡‡‡HA2 defining substitution: V176I

§§§S/45 viruses only possess S144N + N158D + I160K + T328A in common with J.2.4.1 and considered most immunologically relevant

+/− CHO = Gain/loss of glycosylation; HA1 = Hemagglutinin head; RBS = Receptor binding site

the SH 2025 vaccine) (2). The subclade J.2 strain is distinguished from parental “J” strains by N122D (A) (−CHO) and K276E (C) substitutions, with additional S145N (A) distinguishing the chosen J.2 vaccine strain from other J.2 viruses (Table 1, Figure 1A). We also note that the egg-based NH 2025–2026 A/Croatia/10136RV/2023 (subclade J.2) high-growth reassortant strain (X-425A) bears an additional D186A (B) mutation that is frequently associated with egg adaptation and distinguishes it from the 2025–2026 cell-based (and 2024–2025 cell- and egg-based) vaccine strains.

### Historic cluster transitions and associated amino acid changes

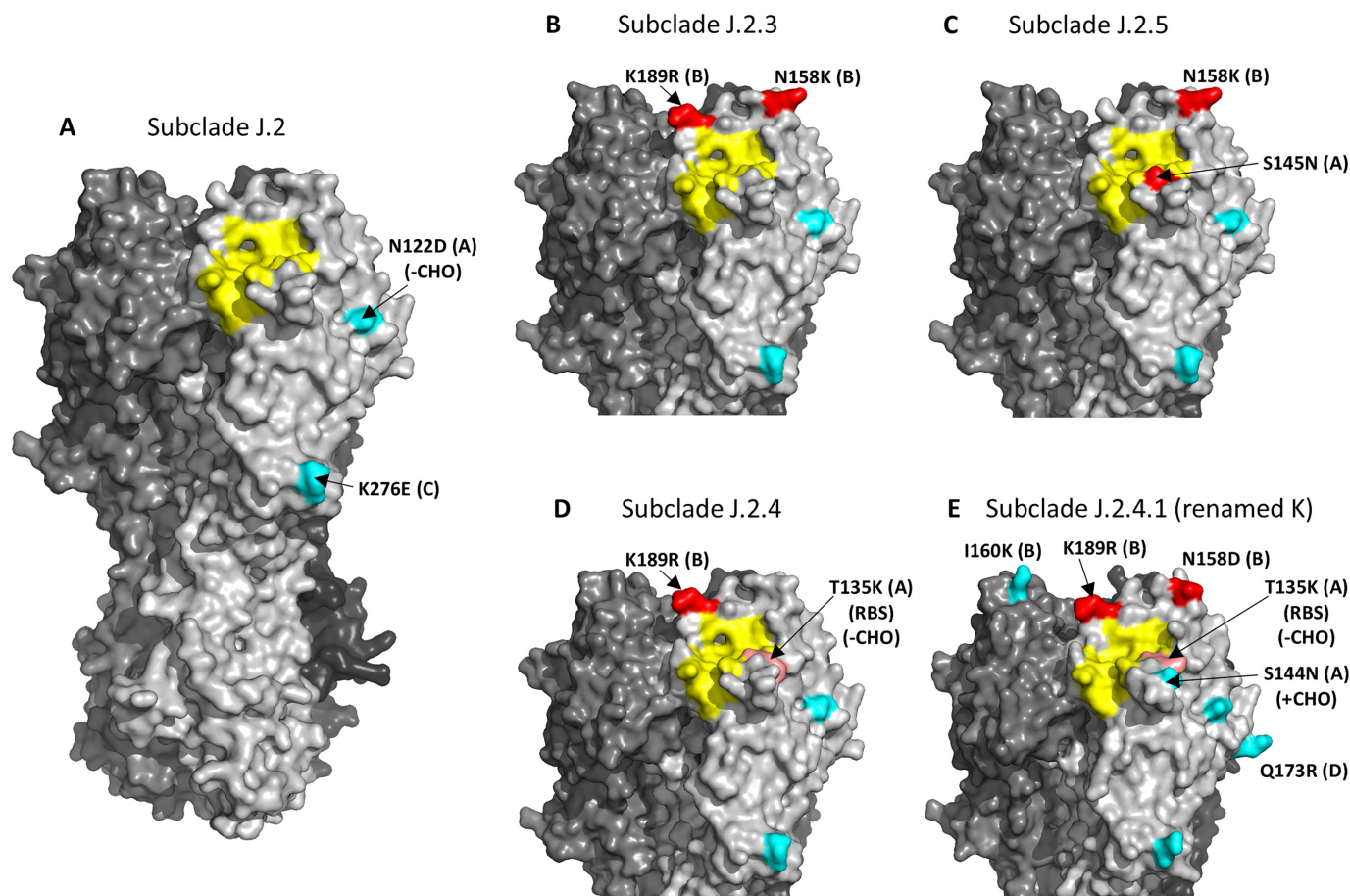
Historic amino acid profiles provide context to anticipate the impact of the same mutations among contemporary viruses and to inform potential cohort effects due to distant childhood imprinting, with implications for pre-existing immunity and/or VE (19). Using >210,000 A(H3N2) HA sequences available in GISAID between January 1, 1968 and August 30, 2025 to generate mutational profiles, we display percent amino acid contribution by year of virus collection at residues affected during the

2024–2025 influenza season and annotate their past involvement in major antigenic cluster transitions requiring vaccine update.

## RESULTS

### Recent global mutational frequencies

**Overview:** Of A(H3N2) viruses collected globally between September 1, 2024 and August 30, 2025, 24,261/24,339 (>99%) were clade 2a.3a.1; subclade contributions are shown in Table 1, with virtually all being J.2 or its descendants (23,982/24,261; 99%). About half (11,790/24,261; 49%) of 2a.3a.1 sequences remained wildtype at all nine antigenic cluster transition and accessory positions. Parental subclade J.2, which has no clade-defining cluster transition site substitutions, was most prevalent (16,960/24,261; 70%), but more than half (9,112/16,960; 54%) had additional mutations at cluster transition sites including several J.2 descendant subclades (eg, J.2.3, J.2.4, J.2.4.1, and J.2.5) defined by two or more such cluster transition site substitutions (Figure 1).



**Figure 1:** Three-dimensional structures of influenza A(H3N2) subclade J.2 variants with mutations at antigenic cluster transition sites

**Note:** Crystal structures of hemagglutinin (HA) trimers, with each identical monomer coloured in a different shade of grey, modelled on A/Victoria/361/2011 (Protein Data Bank accession number 4WE8) in SWISS-MODEL (39), and visualized in PyMOL Molecular Graphics System (version 3.0.3; Schrödinger) (40). The receptor binding site (RBS) is coloured yellow. All substitutions are in relation to the 2024–2025 cell-based vaccine strain A/Massachusetts/18/2022 (2a.3a.1, subclade J; EPI\_ISL\_13897082). Mutations are shaded red if affecting a cluster transition site (eg, 145, 158, 189), pink if affecting an accessory position (eg, 135), and cyan if involving an antigenic site but not a cluster transition or accessory amino acid position (eg, 122, 144, 160, 173, 276). Mutations are only coloured on one monomer as most visible. (Figure 1A) Subclade J.2 reference strain A/Sydney/878/2023 (EPI\_ISL\_18485023) with subclade defining N122D (A) (–CHO) and K276E (C) substitutions. Note that the Northern Hemisphere 2025–2026 and Southern Hemisphere 2025 subclade J.2 vaccine strain (cell-based and egg-based) includes additional S145N (A) antigenic cluster site substitution that is not shown and that further distinguishes it from the displayed subclade J.2.3, J.2.4, and J.2.4.1 (recently renamed subclade K), but not J.2.5 viruses. Moreover, the egg-based version of the J.2 vaccine strain includes an additional D186A (B) (noncluster transition site) substitution that is also not shown; (Figure 1B) Subclade J.2.3 reference strain A/Colombia/1851/2024 (EPI\_ISL\_19727802) with defining N158K (B) and K189R (B) cluster transition site substitutions. (Figure 1C) Subclade J.2.5 reference strain A/Arizona/20/2025 (EPI\_ISL\_19759244) with defining S145N (A) and N158K (B) cluster transition site substitutions. (Figure 1D) Subclade J.2.4 reference strain A/Qatar/78253/2024 (EPI\_ISL\_19644045) with defining T135K (A) (RBS) (–CHO) and K189R (B) accessory and antigenic cluster transition site substitutions, respectively; in September 2025, the World Health Organization recommended a representative J.2.4 strain for the updated 2026 Southern Hemisphere vaccine. (Figure 1E) Further drifted subclade J.2.4.1 reference strain A/Wisconsin/114/2025 (EPI\_ISL\_20131564), recently renamed subclade K, with the same T135K (A) (RBS) (–CHO) and K189R (B) substitutions clade-defining for all J.2.4, plus additional N158D (B) cluster transition site plus S144N (A) (+CHO), T160I (B), and Q173R (D) noncluster transition site substitutions +/- CHO = Gain/loss of glycosylation; RBS = Receptor binding site

Accessory position 135 was mutated in one-quarter of 2a.3a.1 sequences (6,056/24,261; 25%), mostly monophyletically as an additional non-subclade-defining T135K/A (A) (RBS) (−CHO) substitution among predominant J.2 viruses (5,608/6,056; 93%) and also as a subclade-defining substitution of J.2.4 (354/6,056; 6%). Cluster transition residue 145 was also mutated in about one-quarter of 2a.3a.1 sequences (6,886/24,261; 28%), with S145N (A) instead notably accrued as parallel substitution in multiple different subclades, including J.1.1 (subclade defining), J.2 (including the 2025–2026 vaccine reference strain), J.2.1, J.2.2, and J.2.5 (subclade defining).

Cluster transition residues 158 and 189 were mutated in 5%–10% of 2a.3a.1 sequences overall globally across the analysis period, notably including N158K (B) (1,743/24,261; 7%) and K189R (B) (1,156/24,261; 5%), which together were subclade defining for recently emerging J.2.3 (N158K + K189R) viruses (501/24,261; 2%). Both N158K and K189R were also found as parallel and subclade-defining substitutions of emerging J.2.4 (T135K + K189R) (354/24,261; 1%) and J.2.5 (S145N + N158K) (1,197/24,261; 5%) viruses.

Of the 354 J.2.4 viruses, 45 (13%) are noteworthy for having further drifted to accrue a third cluster transition substitution additional to T135K + K189R, namely N158D (B), as well as S144N (A) (+CHO), I160K (B), and Q173R (D) antigenic site substitutions. This further drifted J.2.4.1 variant recently re-labelled subclade K by Nextstrain (17), bears triple cluster transition site mutations (T135K + K189R + N158D). This triply mutated subclade K was identified among available sequences from June (week 26) through end of the analysis period (week 35) of 2025 from all continents except Africa and South America. Other cluster transition site substitutions or combination of substitutions were found in <1% of global sequences.

**2025–2026 northern hemisphere temporal trends:** Given so few sequences available from elsewhere ( $n = 1,268$ , mostly the Middle East), we restrict our analysis of temporal NH trends to sequences from Canada ( $n = 1,598$ ), the United States (US) ( $n = 8,915$ ), and Europe ( $n = 8,268$ ). The combined analysis for North America ( $n = 10,513$ ) likewise excludes Mexico ( $n = 88$ ).

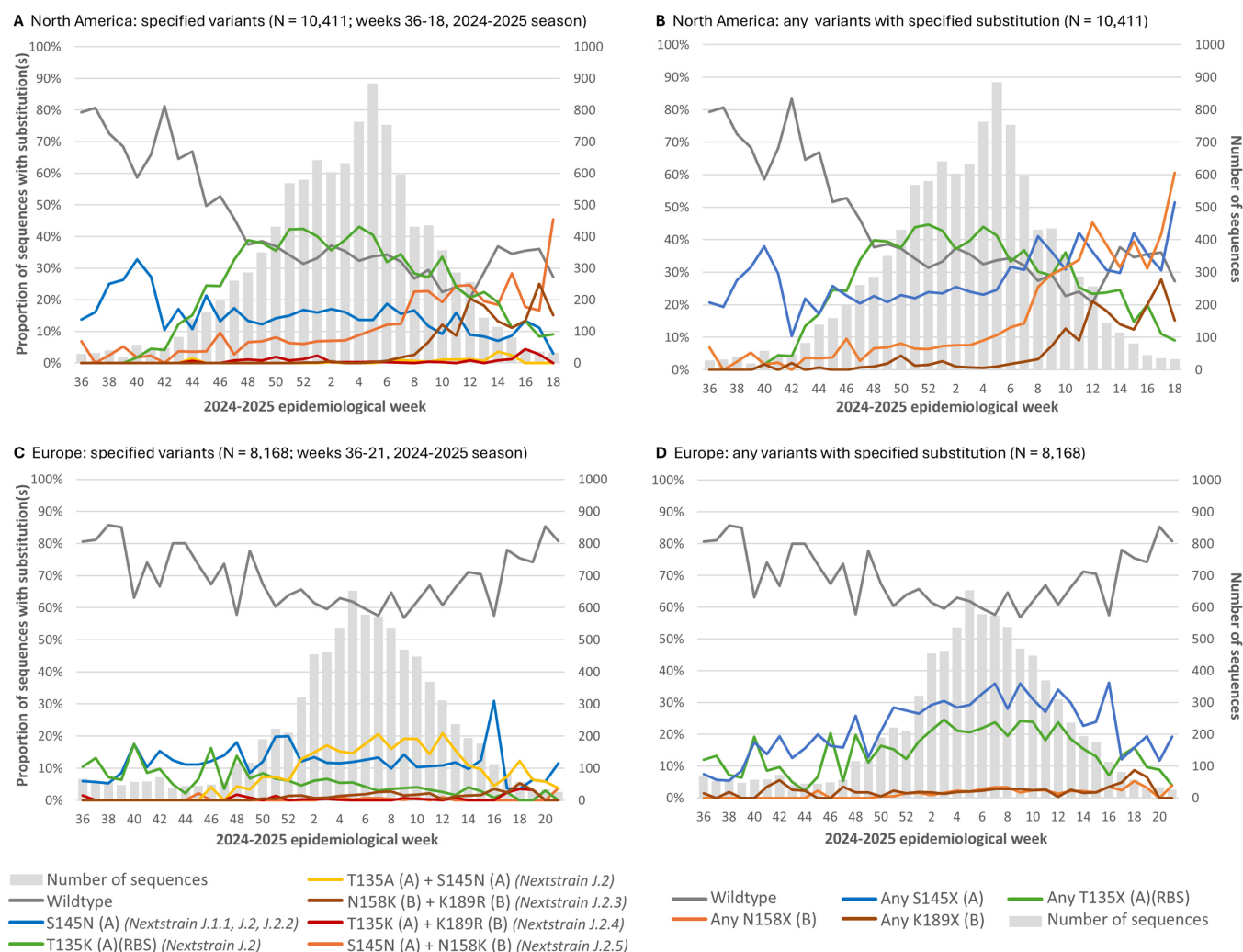
Within North America, wildtype 2a.3a.1 viruses comprised just over one-third of sequences (3,697/10,513; 35%) between September 2024 and August 2025. A decrease in wildtype viruses mirrored an increase in J.2 variants bearing an additional (non-clade-defining) T135K substitution between weeks 41 and 48 (Figure 2A). Subsequently, wildtype and T135K viruses comparably contributed about 30%–40% of J.2 viruses through week 8, with wildtype increasing

thereafter as T135K subsided below 20% by week 14. As a single substitution, S145N comprised about 10%–20% of weekly sequences (Figure 2A), but as a parallel substitution appearing in multiple subclades and alongside other substitutions, viruses with any position 145 substitution increased across the season to 40% or more, mostly reflecting mid-season emergence of J.2.5 (S145N + N158K) (Figure 2B). In that regard also, position 158 parallel substitutions appearing in both J.2.3 and J.2.5 increased to 40% or more of sequenced viruses by end of the season, comparable to any position 145 mutation (Figure 2B). Representing a negligible proportion through week 8, J.2.3 (N158K + K189R) increased to about 10%–20% of weekly sequences through week 18, whereas J.2.4 (T135K + K189R) contributed negligibly throughout. Of the 45 subclade K (T135K + K189R + N158D) viruses identified globally across the analysis period, 7 (16%) were observed in North America with collection dates spanning June 23 (week 26) to August 6 (week 32), 2025.

Restricted to Canada alone, wildtype 2a.3a.1 viruses comprised about half of sequences overall (785/1,598; 49%) between September 2024 and August 2025 (Figure 3A). Temporal patterns were like those in North America overall for the T135K and S145N contribution, including a notable mid-season increase in J.2.5 from week 8. Recognizing reduced sample size, end-of-season contribution was slightly less pronounced for J.2.3 and more for J.2.4 (none the further drifted subclade K) relative to the US contribution. A mix of circulating variants with cluster transition site and/or accessory substitutions was identified in each province (Figure 3B), but with small sample sizes for some provinces, geographic differences are difficult to interpret.

Within Europe, wildtype 2a.3a.1 viruses comprised nearly two-thirds of sequences overall (5,278/8,264; 64%) between September 2024 and August 2025, higher than North America (Figure 2C). Also differing from North America, variants with T135K contributed only minimally, whereas T135A + S145N, as additional substitutions among J.2 parental strains, increased to about 20% per week, surpassing the S145N contribution alone (10%–20%) during peak weeks. Considering parallel substitutions affecting several subclades, the percent of sequences with any position 145 mutation slightly exceeded, but was foremost driven by, variants additionally bearing the position T135A substitution (Figure 2D). Despite a late-season increase in North America, the J.2.3 (N158K + K189R), J.2.4 (T135K + K189R), and J.2.5 (S145N + N158K) variants contributed only negligibly in Europe, with J.2.3 mostly identified in the Netherlands, where it comprised 14% (103/740) of sequences. Of the 45 subclade K (T135K + K189R +





**Figure 2:** Temporal distribution of influenza A(H3N2) clade 2a.3a.1 variants, 2024–2025 season

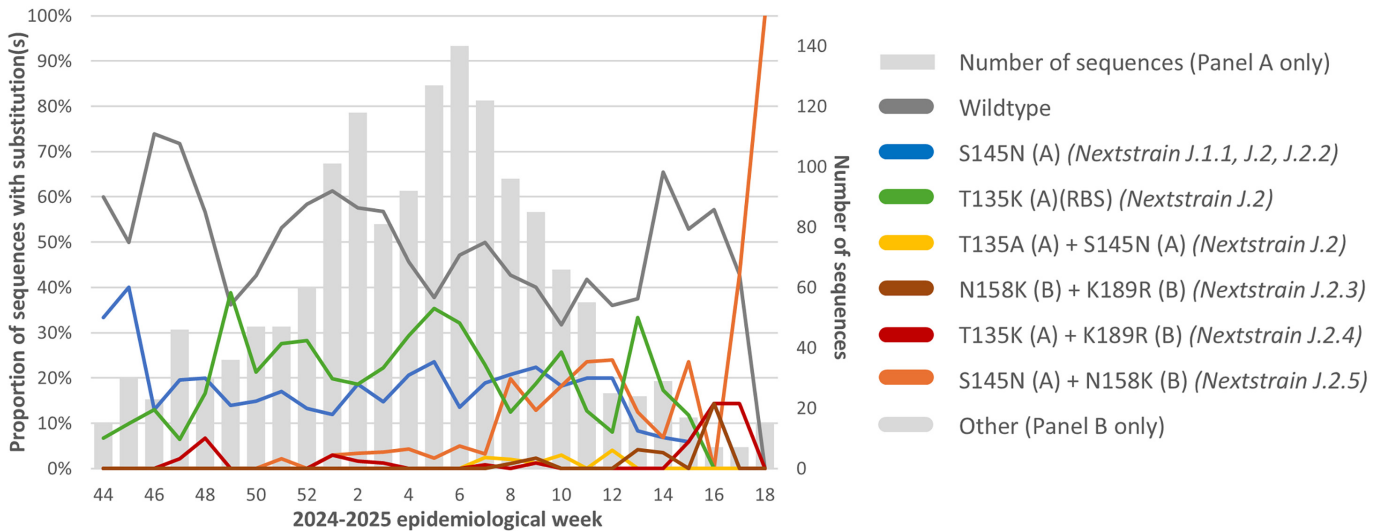
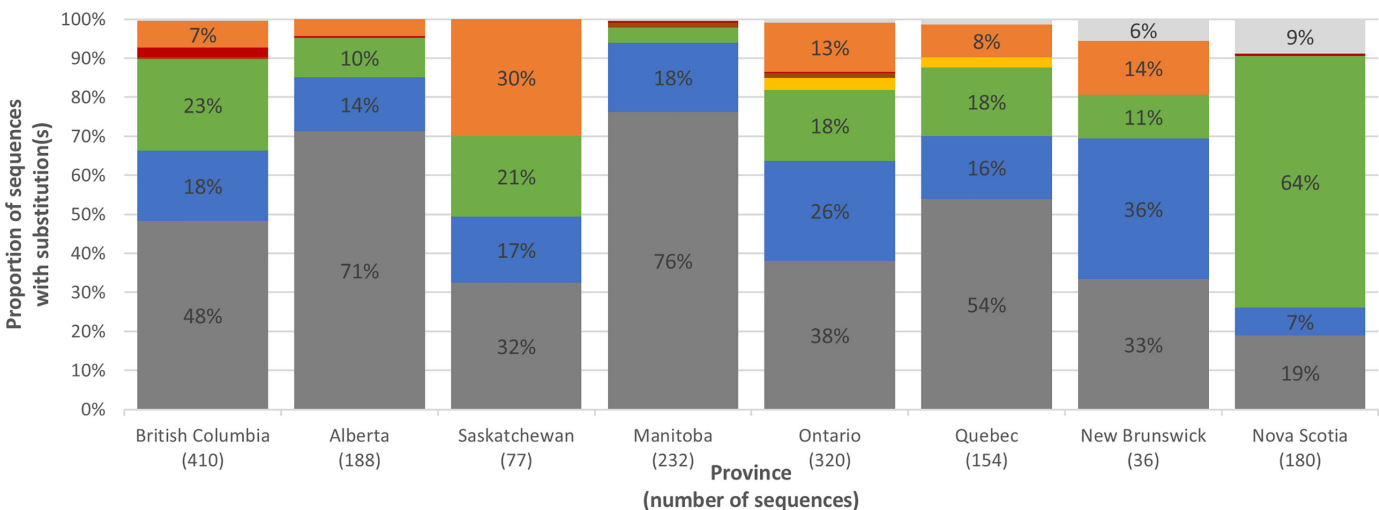
*Note:* Percent of all influenza A(H3N2) clade 2a.3a.1 sequences within the Global Initiative on Sharing All Influenza Data (GISAID) (14) by epidemiological week in North America (Canada and United States only; N = 10,411; weeks 36–18) and in Europe (N = 8,168; weeks 36–21) with the specified substitutions at antigenic cluster transition residues 135, 145, 158, and/or 189, relative to the 2024–2025 Northern Hemisphere influenza A(H3N2) cell-passaged clade 2a.3a.1 (subclade J) vaccine strain (A/Massachusetts/18/2022). (Figure 2A) Specified variants with annotated substitutions restricted to North America. (Figure 2B) Any variants with specified substitutions restricted to North America. (Figure 2C) Specified variants with annotated substitutions restricted to Europe. (Figure 2D) Any variants with specified substitutions restricted to Europe. Data only displayed for weeks with >20 viral sequences and variants present in >1% of global sequences. Wildtype refers to sequences with no mutations at any of the nine antigenic cluster transition and accessory sites relative to the cell-passaged A/Massachusetts/18/2022 anchor strain (ie, N133, T135, S145, T155, S156, N158, N159, K189, and S193). T135X includes T135A/K and S145X includes S145N/G, while N158X includes almost exclusively N158K and K189X almost exclusively K189R

N158D) viruses for which sequences were available globally across the analysis period, 12 (27%) were identified in Europe between July 26 (week 30) and August 25 (week 35), 2025.

**2025 southern hemisphere observations:** During the SH 2025 influenza season, the A(H1N1)pdm09 subtype predominated overall, with A(H3N2) making a substantial

contribution within Africa and Southeast Asia (1,3). From week 1 through week 35 of 2025, all A(H3N2) viruses in Australia belonged to J.2 or its descendants, with <40% (132/340; 39%) remaining wildtype. Subclade J.2.4 (T135K + K189R), which made only a negligible contribution during the NH 2024–2025 season, instead showed a noteworthy increase in Australia from <5% of sequences from weeks 16–25 (2/55; 4%) to nearly 50%



**A** Canada: specified variants (N = 1,541; weeks 44-18, 2024-2025 season)**B** Canada: percent contribution of specified variants within each province (N = 1,597; weeks 36-35, 2024-2025 season)**Figure 3:** Temporal and provincial distribution of influenza A(H3N2) clade 2a.3a.1 variants, Canada, 2024–2025 season

*Note:* Percent of all Canadian influenza A(H3N2) clade 2a.3a.1 sequences within the Global Initiative on Sharing All Influenza Data (GISAID) (14) by epidemiological week or province with specified substitutions at antigenic cluster transition residues 135, 145, 158, and/or 189, relative to the 2024–2025 Northern Hemisphere influenza A(H3N2) cell-passaged clade 2a.3a.1 (subclade J) vaccine strain (A/Massachusetts/18/2022). (Figure 3A) Percent of sequences that were the specified variants, by week. (Figure 3B) Percent of sequences that were the specified variants, by province. Figure 3A restricted to the period spanning October 27, 2024, to May 4, 2025 (weeks 44–18) ( $n = 1,541$ ), owing to few sequences outside that range. Figure 3B includes all Canadian A(H3N2) sequences available between September 1, 2024, and August 30, 2025 (weeks 36–35;  $n = 1,597$ ), excluding Newfoundland due to low sample size ( $n = 1$ ; J.2 + T135K) and other provinces/territories with no publicly available sequences. Note that no available sequences in Canada between weeks 36 of 2024 and 35 of 2025 were the further drifted J.2.4.1 variant

from weeks 31–35 (20/44; 45%). During the same period in South America, all A(H3N2) viruses belonged to J.2 or its descendants, with a similar proportion (172/468; 37%) remaining wildtype but with the doubly mutated J.2.3 (N158K + K189R) instead showing a noteworthy

increase from <10% in weeks 11–20 (12/174; 7%) to nearly two-thirds in weeks 26–35 (49/75; 65%).

In Africa, nearly two-thirds of 2a.3a.1 sequences since week 1 remained wildtype (617/990; 62%), with J.2 or its descendants also predominating (939/990; 95%). Among

J.2 strains, few had additional S145N (91/990; 9%) or the combined T135A + S145N substitutions (15/990; 12%) that were identified in Europe. Of note, within Zambia, J.2.4 (T135K + K189R) predominated (51/63; 81%), and in South Africa, another cluster transition substitution, N159S (B), only sporadically detected in the NH, comprised 10% of 2a.3a.1 sequences (67/699). Finally, in Southeast Asia, about one-third of sequences remained wildtype (185/527; 35%), again almost exclusively J.2 or its descendants (524/527; 99%), with a substantial proportion bearing additional S145N (225/527; 43%) but fewer with T135K (71/527; 13%) compared to North America and with a sudden dramatic increase in the J.2.4 contribution from 1% in weeks 21–30 (1/86) to 23% in weeks 31–35 (14/61).

Of the 45 subclade K viruses identified globally across the September 2024 to August 2025 analysis period, 24 (53%) were identified in the SH between July 17 (week 29) and August 26 (week 35), 2025. The contribution by J.2.5 viruses that was observed in North America was not sustained in the SH.

### Antigenic characterization summaries

Relative to the NH 2024–2025 cell-based (subclade J) vaccine, publicly available antigenic characterization data show most J.2 viruses remained antigenically similar (<8-fold HI titre reduction), including wildtype J.2 (178/253; 70%) and descendants with additional S145N (33/58; 57%), T135K (20/30; 67%) or T135A + S145N (5/6) substitutions (Table 2) (13). However, J.2 viruses that were N158K ( $n = 1$ ) or K189R ( $n = 4$ ) mutated, as well as doubly mutated J.2.3 (N158K + K189R) ( $n = 10$ ), J.2.4 (T135K + K189R) ( $n = 1$ ), and J.2.5 (S145N + N158K) ( $n = 1$ ) viruses, were antigenically distinct from the NH 2024–2025 cell-based vaccine, with similar findings relative to the egg-adapted subclade J vaccine strain.

Relative to the NH 2025–2026 cell-based (subclade J.2) vaccine, available antigenic characterization data show overall improvement for J.2 variants, but this was less evident for the doubly mutated J.2.3 (N158K + K189R) variant (Table 2) (13). In relation to the egg-adapted version of the J.2 vaccine reference strain, more test viruses overall were antigenically distinct at the level of 8-fold reduction, with the doubly mutated J.2.3 viruses exhibiting  $\geq 16$ -fold reduction, including virtually all with  $\geq 32$ -fold titre reduction (15/16; 94%). Most doubly mutated J.2.4 (T135K + K189R) viruses showed  $\geq 16$ -fold HI titre reduction (10/17; 59%) and all subclade K viruses bearing triple 135, 189, and 158 substitutions showed 32-fold reductions, but this was based on few test viruses (3/3) (3). We note as potentially contributory that the selected 2025–2026 NH J.2

reference vaccine strain (cell- and egg-based) is N145 and, in that regard, differs from other J.2 viruses, including the J.2.3, J.2.4, and K variants that remain wildtype S145; moreover, the egg-based version of the NH 2025–2026 vaccine bears an additional D186A egg-adaptation mutation that distinguishes it from circulating strains more generally.

### Historic amino acid profiles

Drawing upon 210,881 sequences collected between January 1, 1968 and August 30, 2025, we display percent amino acid contribution by year at residues 135, 145, 158, and 189 (Figure 4). We annotate their past direct involvement in the 18 major antigenic cluster transitions between 1968 and 2018 (20).

Of at least three historical amino acid transitions at accessory position 135 (Figure 4A), just one facilitated major antigenic cluster transition, namely K135T (+CHO) in association with E158K (+ K156Q) in the transition from the variant cluster represented by A/Wuhan/359/1995 (WU95) to the cluster represented by A/Sydney/5/1997 (SY97). Of at least four historical amino acid transitions at residue 145 (Figure 4B), three were directly involved in major antigenic cluster transitions (1989, 1995, 2004). Note, however, that similar S145N substitution in the transition from A/England/42/1972 (EN72) to A/Victoria/3/1975 (VI75) or reverse N145S substitution from A/Texas/50/2012 (clade 3C.1) to A/Hong Kong/4801/2014 (clade 3C.2a) did not directly contribute to those particular antigenic cluster transitions, which were instead due to concomitant mutations at other antigenic cluster transition residues (ie, Q189K and K160T (+CHO), respectively) (20).

All three historical amino acid transitions at position 158 (Figure 4C) were directly involved in major antigenic cluster transition (1977, 1997, 2009), including prior K158N transition from A/Brisbane/59/2007 (BR07) to A/Victoria/2009/2009 (VI09), reverting in 2024–2025 through the contemporary N158K substitution. Note, however, that each historic antigenic cluster transition involving position 158 also involved mutation at another cluster transition residue (ie, D193N, K156Q, or N189R, respectively) (20). Finally, three of five amino acid transitions at position 189 (Figure 4D) were directly involved in antigenic cluster transitions (1975, 1987, 2009), including the same K189R substitution from A/Bangkok/1/1979 (BK79) to A/Sichuan/2/1987 (SI87), as recently observed in 2024–2025.

## DISCUSSION

After several seasons of only incremental drift, A(H3N2) viruses are showing important mutations at HA residues

**Table 2:** Fold-reduction in hemagglutination inhibition titres of A(H3N2) J.2 variants relative to cell- or egg-based vaccine strains

| 2a.3a.1<br>J.2 test<br>variant<br>(antigenic site or receptor<br>binding site involvement<br>or subclade-defining<br>cluster transition site<br>substitutions) | 2024–2025 A(H3N2) reference vaccine strains*<br>(2a.3a.1; subclade J) |             |            |             |                                                     |             |            |             | 2025–2026 A(H3N2) reference vaccine strains†<br>(subclade J.2) |              |           |            |                                                     |             |            |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------|------------|-------------|-----------------------------------------------------|-------------|------------|-------------|----------------------------------------------------------------|--------------|-----------|------------|-----------------------------------------------------|-------------|------------|-------------|
|                                                                                                                                                                | A/Massachusetts/18/2022<br>(cell-based)                               |             |            |             | A/Thailand/08/2022<br>(egg-based)                   |             |            |             | A/District of Columbia/27/2023<br>(cell-based)                 |              |           |            | A/Croatia/10136RV/2023<br>(egg-based)‡              |             |            |             |
|                                                                                                                                                                | <i>n</i> with specified fold-<br>reduction (%; row)                   |             |            |             | <i>n</i> with specified fold-<br>reduction (%; row) |             |            |             | <i>n</i> with specified fold-<br>reduction (%; row)            |              |           |            | <i>n</i> with specified fold-<br>reduction (%; row) |             |            |             |
|                                                                                                                                                                | <b>N</b>                                                              | ≤4          | 8          | ≥16         | <b>N</b>                                            | ≤4          | 8          | ≥16         | <b>N</b>                                                       | ≤4           | 8         | ≥16        | <b>N</b>                                            | ≤4          | 8          | ≥16         |
| Wildtype <sup>§</sup>                                                                                                                                          | <b>253</b>                                                            | 178<br>(70) | 69<br>(27) | 6<br>(2)    | <b>253</b>                                          | 206<br>(81) | 45<br>(18) | 2<br>(1)    | <b>356</b>                                                     | 355<br>(>99) | 1<br>(<1) | 0          | <b>360</b>                                          | 257<br>(71) | 89<br>(25) | 14<br>(4)   |
| S145N (A) <sup>¶</sup>                                                                                                                                         | <b>58</b>                                                             | 33<br>(57)  | 20<br>(34) | 5<br>(9)    | <b>58</b>                                           | 37<br>(64)  | 18<br>(31) | 3<br>(5)    | <b>87</b>                                                      | 87<br>(100)  | 0         | 0          | <b>91</b>                                           | 76<br>(84)  | 15<br>(16) | 0           |
| T135K (A) (RBS) <sup>**</sup>                                                                                                                                  | <b>30</b>                                                             | 20<br>(67)  | 8<br>(27)  | 2<br>(7)    | <b>30</b>                                           | 21<br>(70)  | 8<br>(27)  | 1<br>(3)    | <b>39</b>                                                      | 39<br>(100)  | 0         | 0          | <b>40</b>                                           | 30<br>(75)  | 10<br>(25) | 0           |
| T135A + S145N                                                                                                                                                  | <b>6</b>                                                              | 5           | 0          | 1           | <b>6</b>                                            | 2           | 4          | 0           | <b>17</b>                                                      | 17<br>(100)  | 0         | 0          | <b>19</b>                                           | 17<br>(89)  | 2<br>(11)  | 0           |
| N158K (B)                                                                                                                                                      | <b>1</b>                                                              | 0           | 0          | 1           | <b>1</b>                                            | 0           | 0          | 1           | <b>1</b>                                                       | 0            | 1         | 0          | <b>1</b>                                            | 0           | 1          | 0           |
| K189R (B)                                                                                                                                                      | <b>4</b>                                                              | 0           | 2          | 2           | <b>4</b>                                            | 0           | 2          | 2           | <b>4</b>                                                       | 4            | 0         | 0          | <b>4</b>                                            | 0           | 0          | 4           |
| J.2.3 (N158K + K189R) <sup>††</sup>                                                                                                                            | <b>10</b>                                                             | 0           | 0          | 10<br>(100) | <b>10</b>                                           | 0           | 0          | 10<br>(100) | <b>14</b>                                                      | 0            | 2<br>(14) | 12<br>(86) | <b>16</b>                                           | 0           | 0          | 16<br>(100) |
| J.2.4 (T135K + K189R)                                                                                                                                          | <b>1</b>                                                              | 0           | 0          | 1           | <b>1</b>                                            | 0           | 0          | 1           | <b>13</b>                                                      | 8<br>(62)    | 4<br>(31) | 1<br>(8)   | <b>17</b>                                           | 1<br>(6)    | 6<br>(35)  | 10<br>(59)  |
| J.2.4.1 (now subclade K)<br>(T135K + K189R +<br>N158D) <sup>††</sup>                                                                                           | <b>0</b>                                                              | –           | –          | –           | <b>0</b>                                            | –           | –          | –           | <b>0</b>                                                       | –            | –         | –          | <b>3</b>                                            | 0           | 0          | 3           |
| J.2.5 (S145N + N158K)                                                                                                                                          | <b>1</b>                                                              | 0           | 0          | 1           | <b>1</b>                                            | 0           | 0          | 1           | <b>9</b>                                                       | 9            | 0         | 0          | <b>9</b>                                            | 6           | 3          | 0           |

**Table 2:** Continued (abbreviations, footnotes)

All substitutions are in relation to the cell-based A/Massachusetts/18/2012 vaccine reference strain (clade 2a.3a.1, subclade J). HI findings shown by individual antigenic cluster transition site substitutions were verified by corresponding sequence data within the Global Initiative on Sharing All Influenza Data (GISAID) (14). Percentages are displayed only when at least 10 test viruses are included in antigenic characterization. Antigenic distinction relative to the specified vaccine reference strain is defined by 8-fold or greater reduction in HI titre (18). Summarized HI findings include results extracted from the Crick Institute February and September 2025 interim reports for 364 viruses tested against all four vaccine reference strains plus 176 viruses tested only against the 2025–2026 cell- and egg-based vaccine reference viruses (13); additionally, we include 20 unique results extracted from the September 2025 report of the WHO SH 2026 vaccine strain selection meeting for viruses tested relative to the 2025–2026 egg-based vaccine reference virus only (3)

\*Influenza A(H3N2) vaccine reference strains for Northern Hemisphere 2024–2025 and Southern Hemisphere 2024 influenza season

†Influenza A(H3N2) vaccine reference strains for Northern Hemisphere 2025–2026 and Southern Hemisphere 2025 influenza season. Note that the J.2 vaccine strain (cell- and egg-based) is N145, which differs from other J.2 viruses, including the J.2.3, J.2.4, and J.2.4.1 (the latter recently renamed subclade K) variants that remain wildtype S145

‡The NH 2025–2026 egg-based A/Croatia/10136RV/2023 (subclade J.2) high-growth reassortant strain (X-425A) bears an additional D186A (B) mutation that is frequently associated with egg adaptation and distinguishes it from the 2025–2026 cell-based and 2024–2025 cell- and egg-based vaccine strains

§Wildtype viruses (defined by no mutations in any of the nine cluster transition or accessory positions) include J.2, J.2.1, and J.2.2 subclades

¶S145N viruses include mutated J.2 and J.2.2 subclade viruses

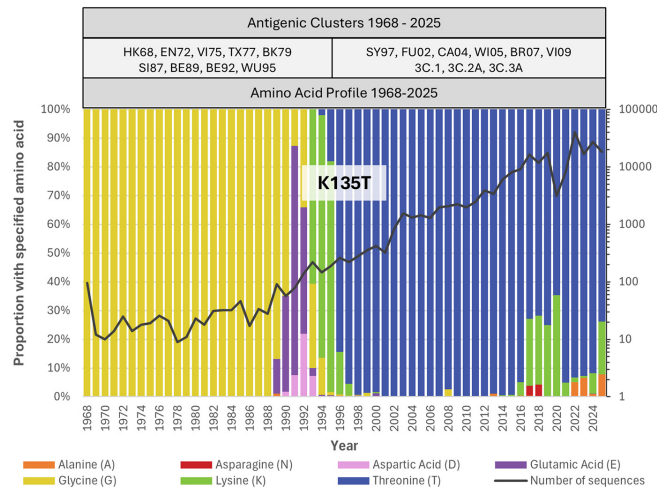
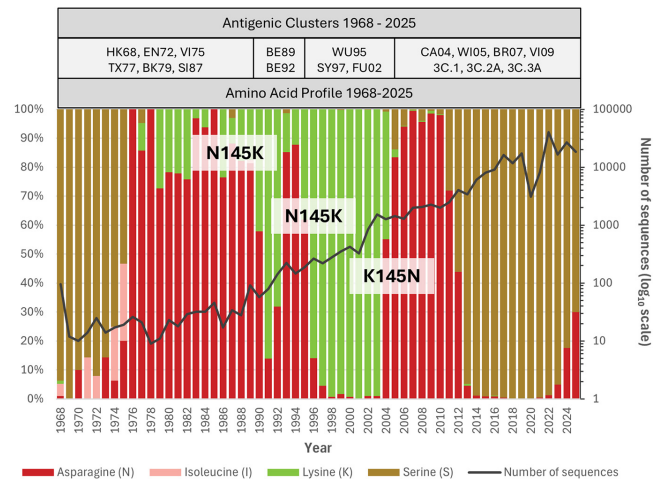
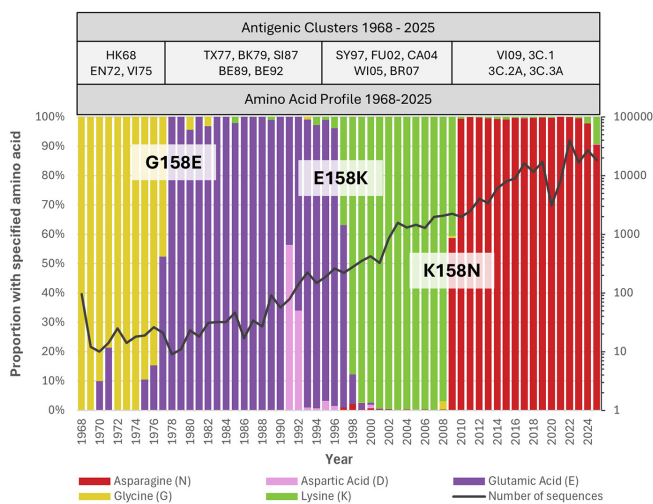
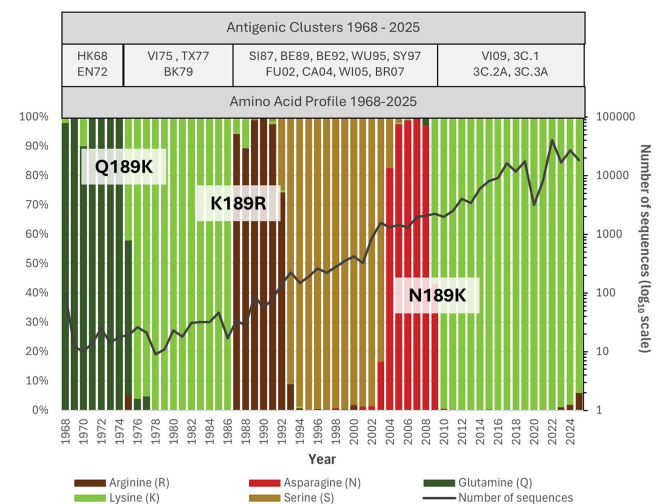
\*\*T135K viruses include mutated J.2 and J.2.1 subclade viruses

††Most J.2.3 viruses were not recognized by any of the antisera (below detectable level of <40 HI units), which, relative to the displayed cell-based 2024–2025 A/Massachusetts/18/2022 and 2025–2026 A/District of Columbia/27/2023 vaccine strains signifies  $\geq 16$ -fold reduction, relative to the egg-based 2024–2025 A/Thailand/08/2022 vaccine strain signifies  $\geq 32$ -fold reduction, and relative to the egg-based 2025–2026 A/Croatia/10136RV/2023 vaccine strain signifies  $\geq 64$ -fold titre reduction

‡‡Additional antigenic site substitutions within this further drifted J.2.4.1 variant (T135K, N158D, K189R), now called subclade K, include S144N (A) (+CHO), I160K (B), and Q173R (D). All 3/3 show 32-fold reduction in HI titre relative to the 2025–2026 egg-based A/Croatia/10136RV/2023 vaccine strain. No antigenic characterization data are publicly available to date for subclade K viruses relative to the cell-based A/District of Columbia/27/2023 vaccine strain (as of October 17, 2025)

HI = Hemagglutination inhibition; RBS = Receptor binding site



**A** Residue 135 (Antigenic site A)**B** Residue 145 (Antigenic site A)**C** Residue 158 (Antigenic site B)**D** Residue 189 (Antigenic site B)

**Figure 4:** Percent amino acid contribution yearly at influenza A(H3N2) antigenic cluster transition sites, worldwide by year of collection, 1968 to 2025 (N = 210,881)

*Note: The percent yearly contribution by specified amino acids at influenza A(H3N2) hemagglutinin head antigenic cluster transition or accessory residues 135 (Figure 4A), 145 (Figure 4B), 158 (Figure 4C), and 189 (Figure 4D). The included number of A(H3) sequences by year is shown as the dark grey line on the secondary y-axis (log<sub>10</sub> scale). Infrequent substitutions (never surpassing 1% of yearly totals) are excluded. Strains representing major antigenic clusters are overlaid above the amino acid profile of each residue, grouped together based on direct involvement in transitions between clusters (7,8,10,11,20). Historically significant substitutions responsible for each cluster transition are annotated in white boxes within the figure profiles. List of cluster abbreviations: HK68 = A/Hong Kong/1/1968; EN72 = A/England/42/1972; VI75 = A/Victoria/3/1975; TX77 = A/Texas/1/1977; BK79 = A/Bangkok/1/1979; SI87 = A/Sichuan/2/1987; BE89 = A/Beijing/353/1989; BE92 = A/Beijing/32/1992; WU95 = A/Wuhan/359/1995; SY97 = A/Sydney/5/1997; FU02 = A/Fujian/411/2002; CA04 = A/California/7/2004; WI05 = A/Wisconsin/67/2005; BR07 = A/Brisbane/59/2007; VI09 = A/Victoria/208/2009; 3C.1 = A/Texas/50/2012; 3C.2a = A/Hong Kong/4801/2014; 3C.3a = A/Kansas/14/2017*

historically involved in major antigenic cluster transitions, namely parallel substitutions affecting residues 135, 145, 158, and/or 189, alone or in combination. Although the WHO updated from the subclade J strain for the NH 2024–

2025 (and SH 2024) vaccine to a subclade J.2 strain for the NH 2025–2026 (and SH 2025) vaccines (2), the circulating J.2 subclade has continued to diversify such that the WHO recently recommended a further update for the

SH 2026 vaccine to a J.2.4 strain (3). Accrual of the same substitutions among different J.2 descendants across large geographic regions suggests the mutations confer fitness advantage and immune escape potential. This notably includes variants with a combination of cluster transition mutations such as J.2.3 (N158K + K189R) and J.2.4 (T135K + K189R) toward the tail-end of the NH 2024–2025 season, sustained and with further J.2.4.1 drift additionally acquiring N158D and other mutations (including glycosylation effects) during the SH 2025 season, warranting recent relabelling as subclade K (17). The immunological landscapes and transmission dynamics underpinning the emergence and relative contribution of immune escape variants may be complex, but recent A(H3N2) observations are sufficiently concerning to warrant close collective monitoring leading into the NH 2025–2026 influenza season.

Historically, amino acid substitutions responsible for large ADs between variant clusters have involved substantial change in biophysical properties affecting protein folding, stability, solubility, and overall antigen–antibody binding, such as electrostatic charge, polarity, length, and/or accessible surface area of the amino acid side chain (20). Glycosylation effects, such as the recent antigenic site A substitutions N122D (–CHO), subclade-defining for J.2, or T135K (–CHO), further subclade defining for J.2.4, or S144N (+CHO) accrued among further drifted J.2.4.1 (now subclade K) viruses, may not directly alter antigenicity but can assist immune evasion through shielding (+CHO) or exposing (–CHO) pivotal viral epitopes (21). A good historical example includes the cluster transition WU95 to SY97 for which cluster transition substitutions E158K + K156Q directly contributed 3.6 units of AD change (>8-fold titre reduction), which increased to 5.5 units (>32-fold reduction) through additional gain of glycosylation due to the accessory K135T (+CHO) substitution (8). In conjunction with N122D (–CHO), contemporary T135K/A (–CHO) loss of glycosylation among some J.2 viruses exposes epitopes within antigenic site A that have been masked since 1997 (22), but with limited independent impact on antigenicity relative to wildtype J.2 viruses (Table 2) (13). Considering its limited historic involvement as an accessory site mediated through glycosylation effects, residue 135 substitutions, including recent T135K/A loss of glycosylation, are most likely to exert an impact in conjunction with other residue changes.

Despite its direct involvement in historical antigenic cluster transitions in 1989, 1995, and 2004, the 145 residue in antigenic site A showing S145N parallel substitution in multiple 2024–2025 subclades is also not expected to confer major antigenic change on its own. This is reinforced by historical observations and the recent antigenicity data we

summarize (Table 2) (13). While S145N and the reverse N145S substitutions have occurred previously (ie, VI75 and 3C.2a emergence, respectively), these particular substitutions involve a switch between two small polar amino acids (serine and asparagine) previously associated with <1 AD unit of change (<2-fold HI titre reduction) (7,8). As such, prior antigenic cluster transitions in the presence of S145N or N145S have been attributed to concomitant change at other cluster transition positions. Such considerations may be relevant in the context of the 2025–2026 NH vaccine being a J.2 strain with S145N substitution, further distinguishing it from the already multiply-mutated variants such as J.2.3, J.2.4, and subclade K, each of which remains wildtype S145.

We draw particular attention to the parallel substitutions affecting antigenic site B residues 158 and 189, associated with greater antigenic distinction. The recent N158K substitution reverts the small polar (hydrophilic) neutral asparagine (N) back to the large positively charged lysine (K) residue last present in BR07, thereby changing the electrostatic charge at the protein surface and increasing the size of the amino acid side chain. While the recent K189R substitution exchanges one large positively charged residue for another, suggesting little antigenic impact, the same K189R substitution in 1987 resulted in an AD change of 3.3 units (>8-fold HI titre reduction) with allosteric effects (eg, side-chain flexibility) likely playing a role (8,20). Deep mutational scanning of human sera collected in 2020 confirms that while T135K and S145N mutations confer no major immunological escape, any mutations at residue 189 should be viewed as immuno-evasive, especially in children (23). Although amino acid 158 mutants could not be rescued for deep mutational scanning, preliminary data from high-throughput sequencing-based neutralization assays show that across multiple subclades, strains with mutations at sites 158 and/or 189 react poorly with sera collected as recently as spring 2025 (24). On that basis, the emergence of subclades with dual substitutions involving positions 158 and 189 seems particularly important to monitor for the coming season, including J.2.3 (N158K + K189R) and the triply mutated subclade K (T135K + N158D + K189R).

While still pending peer-reviewed publication, 2024–2025 end-of-season VE estimates from the Canadian Sentinel Practitioner Surveillance Network (SPSN) using the clade J vaccine strain suggest comparable A(H3N2) variant-specific protection against wildtype, and singly T135K or S145N mutated viruses, but negligible protection against the emergent J.2.5 (S145N + N158K) strain, consistent with antigenicity observations. Interim SH 2025 VE estimates based upon the updated J.2 vaccine indicate VE of

<40% against A(H3N2) hospitalization, but genetic characterization data only specified that all A(H3N2) sequences in contributing countries were clade 2a.3a.1 (25). In its September 2025 report for the SH 2026 vaccine strain selection meeting, the WHO concluded that the SH 2025 (and NH 2025–2026) J.2 vaccine poorly recognizes J.2.3, J.2.4, and J.2.5 variants based upon ferret postinfection antisera and adult postimmunization serological analyses (3).

The WHO ultimately chose a J.2.4 (T135K + K189R) strain for the updated SH 2026 vaccine, in part reflecting its rapid spread and diversification within several SH regions; however, it is too late to change the NH 2025–2026 vaccine, and this choice for the 2026 SH vaccine does not reflect the more recent subclade K variant (3). Having correctly predicted in March 2024 that J.2 would dominate among A(H3N2) viruses during the NH 2024–2025 season (26), Nextstrain now projects on its dashboard (as of October 17, 2025) that the further drifted J.2.4.1 variant, now called subclade K, with the additional N158D and S144N (+CHO), I160K, and Q173R substitutions, is likely to predominate during the 2025–2026 season (27). Data from the WHO September 2025 report show that subclade K is associated with 32-fold HI titre reduction relative to the egg-based NH 2025–2026 vaccine, but this is based on few ( $n = 3$ ) characterized viruses in publicly available data to date (3). Additional antigenicity data, including in relation to the cell-based vaccine reference strain, are urgently needed. Unlike N158K viruses against which there may be some preimmunity among people imprinted as children to K158 between 1997 and 2009, D158 has only ever been identified in a limited way in the human population (Figure 4C). Compared to parental J.2.4 viruses defined by T135K (–CHO), subclade K viruses rapidly acquired S144N (+CHO) substitution in the same antigenic site A region. Whether this glycan swapping in conjunction with N158D contributes to the fitness advantage and/or immune escape potential of subclade K remains to be seen.

### Limitations

The main limitation of surveillance analyses using publicly available genomic and antigenic data is not knowing the underlying sampling frame for included specimens, their representativeness, or weighting relative to population considerations of public health relevance (eg, vaccine status, age distribution, clustering). China and Southeast Asia are the greatest contributors to worldwide influenza fluctuations (28), but deposited viral sequences reflecting clinician or researcher interests or capacity may not proportionately represent variant emergence or transmission

dynamics. Some of our analyses, both genetic and antigenic, are limited by small sample sizes. Available antigenic characterization data from the Crick Institute and WHO are based upon genetically characterized reference strains representative of their circulating counterparts with the same genetic identity (3,13); beyond expected assay variability, their greatest limitation is reliance upon ferret antisera that do not reflect the more complex human immunological landscape. We draw upon historical amino acid profiles to understand contemporary mutations, whereas structural and biochemical interactions with adjacent amino acids and immunological landscapes may differ. Like others, we extrapolate molecular, antigenic, and immunogenicity data to anticipate immune escape and impact, recognizing that direct epidemiological assessment is preferable when possible.

Ultimately, variants may appear within an antigenic cluster up to 2 years before the period in which that cluster becomes dominant, and given so much other complexity it is difficult to predict with certainty when or which variant will predominate (7). While cross-hemisphere migration of influenza strains is well established, the SH does not reliably predict NH influenza activity, or vice versa, with viral seeding conditions, social connectedness, and population immunity as influential context (29). A Canadian study identified limited predictive value of the preceding Australian influenza season for Canada (30), and we therefore interpret SH insights only in conjunction with other sources of information, including Nextstrain projections as of mid-October indicating J.2.4.1, now known as subclade K, is likely to predominate among A(H3N2) strains for the coming NH 2025–2026 season (27).

We focus our analysis on A(H3N2), but A(H1N1)pdm09 (and influenza B) could again make a substantial contribution in 2025–2026. The NH 2025–2026 vaccine against A(H1N1)pdm09 is unchanged from the NH 2024–2025 vaccine, whereas the WHO recommended an A(H1N1)pdm09 vaccine strain update for the SH 2026 formulation (3). Although A(H3N2) co-circulated with A(H1N1)pdm09 in 2023–2024 and 2024–2025, Canada last experienced a dominant A(H3N2) epidemic in 2022–2023, the latter beginning and peaking unusually early in late October (week 43) and late November (week 47), respectively (4,31). We note that as of week 41 of the 2025–2026 season, influenza activity in Canada remains at low interseasonal levels (4), but several regions of the NH (eg, United Kingdom, Japan) are already reporting unusually sharp and early upswing in influenza indicators, foremost due to A(H3N2), of which most (>85%) in the United Kingdom are J.2.4.1, now known as subclade K (32,33).



## CONCLUSIONS

In summary, A(H3N2) viruses have recently shown important genetic diversification with meaningful antigenic impact and immune escape. Several variants, notably J.2.3 and J.2.4, have acquired substitutions at more than one cluster transition residue with a further drifted and vaccine-mismatched subclade K variant projected to predominate during the NH 2025–2026 season. Children are generally more immunologically susceptible, socially interconnected, and subject to the highest influenza attack rates (34–36), but diverse immunological histories among adolescents and adults, including differential childhood imprinting, may also influence age-related influenza risk and VE. While a vaccine may provide cross-protection against antigenically mismatched variants, especially important for those at highest risk of severe influenza complications (37), the magnitude of cross-protection requires within-season quantification. Sentinel surveillance platforms efficiently linking genotypic (eg, sequencing), phenotypic (eg, antigenic characterization, human serology), and epidemiological (eg, VE, disease burden) information will be critical for timely risk assessment and response for the coming NH 2025–2026 season, including recommendations for the appropriate use of antivirals and/or other adjunct measures as indicated (38).

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